

OLED Display

04 July 2025 11:15

-> on board, no name mentioned except for 9E3318A1, and from online resources, says it uses the chip SSD1315.

-> In the datasheet, SSD1315, section 6.1.5, page 15, we can find the slave address for I2C, 0x74 is low and 0x7A is high.

Note: Refer section 6.1 for more imp info.

How OLED works:

- The system sends data over a wire. There are 2 kinds of messages : 1. Commands 2.Data
 1. Command: Tell the display what to do, ex. Turn on/off the screen, print in a particular row etc.
 2. Data : the actual info to be displayed.
- OLED screens are made up of tiny pixel, a pixel is the smallest addressable unit in an image.
- Each pixel is controlled individually using row/column addressing.
- OLED has a tiny chip called display controller, which understands the cmd, data, stores each pixel in a memory called Display RAM and decides which pixel should glow.
- The Display RAM is organized in pages.
- The Display RAM is a map of the screen made up of 1s or 0s.
- The memory is divided into rows and columns, 128x64 means, 128 columns, 64 rows(each page = 8 rows, i.e bits) .
- So 8 pages x 8 rows = 64 vertical pixels.
- Each column has 1 byte per page.
- OLED doesn't understand the characters, it only knows pixels (on/off dots).

-> This oled is also compatible with ssd1306, i.e we can also use ssd1306 libraries.

-> Co - Continuation bit

D/C# - Data/Command Selection bit

ACK - Acknowledgement

SA0 - Slave address bit

R/W# - Read / Write Selection bit

S - Start Condition / P - Stop Condition

-> Read GDDRAM section.

-> Display controller receives data from a MCU in series and can output it in parallel.

-> The output can be shown in parallel by controlling the rows and columns of oled matrix

-> Serial data transmission requires time, and the data must be stored somewhere until it can be processed.

->For that purpose the controller has a buffer memory, this buffer memory collects bits transferred from mcu until they form a full frame.

-> we need a total 8192 bits per frame, then this frame can be transmitted through the oled matrix .

-> 1 bit per pixel